

6**B**

For elastic collisions, the relative speed of approach equals relative speed of separation

$$u - 0 = v_2 - v_1$$

$$v_2 = u + v_1 \quad (1)$$

From conservation of linear momentum

$$mu + 0 = mv_1 + 4mv_2 \quad (2)$$

Substituting (1) into (2)

$$mu = mv_1 + 4m(u + v_1)$$

$$5v_1 = -3u$$

$$v_1 = -\frac{3}{5}u$$

7**D**